

PROMETHEUS Patent Drawings

Hybrid Analog-Digital Computing System with Time-Domain Multiplexed Computation

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1 Brief Description of the Drawings

The invention may be better understood with reference to the following figures, which are illustrative and non-limiting:

- **Figure 1:** System-level block diagram of a hybrid analog-digital computing accelerator
- **Figure 2:** Internal architecture of the analog computing accelerator
- **Figure 3:** Time-division multiplexed analog operation timing diagram
- **Figure 4:** TDM coupling of multiple signals into a single analog state
- **Figure 5:** Calibration and measurement feedback loop
- **Figure 6:** Software control interface and register map

2 Figure 1 — System-Level Block Diagram

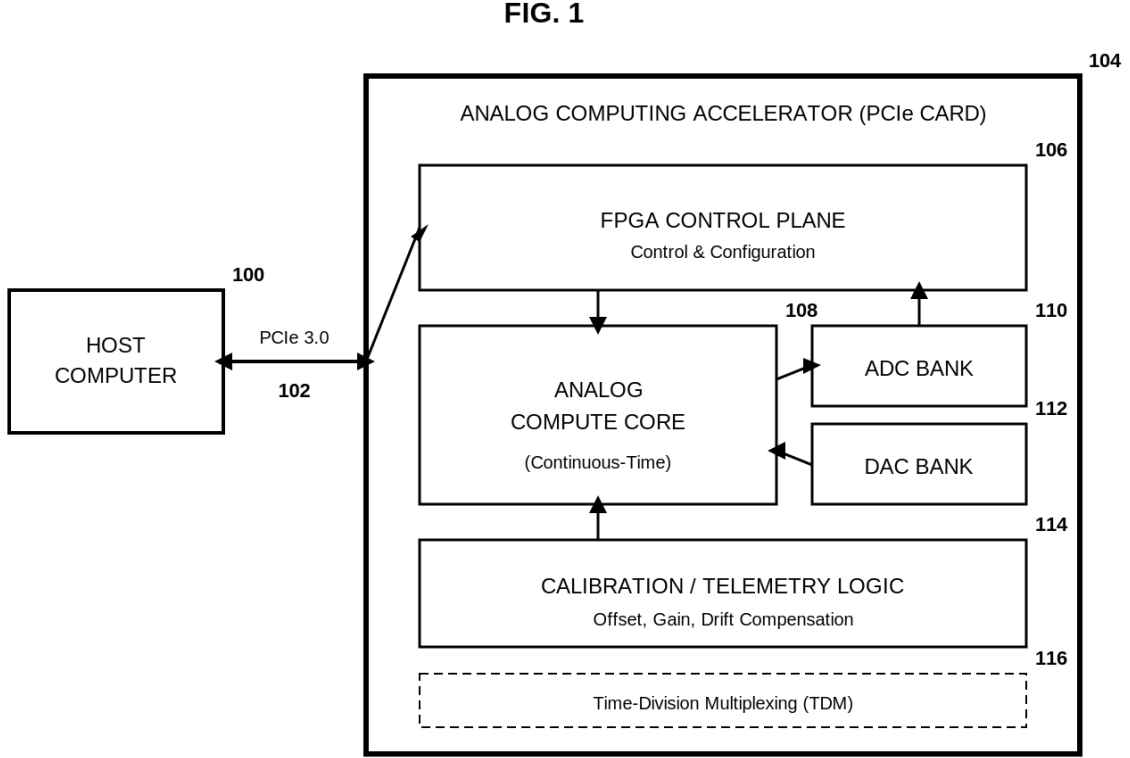


Figure 1: System-Level Block Diagram

2.1 Description of Figure 1

Figure 1 illustrates the overall system architecture of the hybrid analog-digital computing accelerator.

Reference numerals:

- **100** — Host Computer
- **102** — PCIe 3.0 Interface
- **104** — Analog Computing Accelerator (PCIe Card)
- **106** — FPGA Control Plane (Control & Configuration)
- **108** — Analog Compute Core (Continuous-Time)
- **110** — ADC Bank
- **112** — DAC Bank
- **114** — Calibration / Telemetry Logic (Offset, Gain, Drift Compensation)
- **116** — Time-Division Multiplexing (TDM)

The host computer (100) communicates with the analog computing accelerator (104) via a high-speed PCIe 3.0 interface (102). The accelerator comprises an FPGA control plane (106) that supervises and configures the analog compute core (108). The ADC bank (110) and DAC bank (112) provide

conversion between digital and analog domains. A calibration and telemetry logic block (114) maintains accuracy through offset, gain, and drift compensation. The entire analog computation is organized using time-division multiplexing (116).

3 Figure 2 — Internal Architecture of the Analog Computing Accelerator

FIG. 2

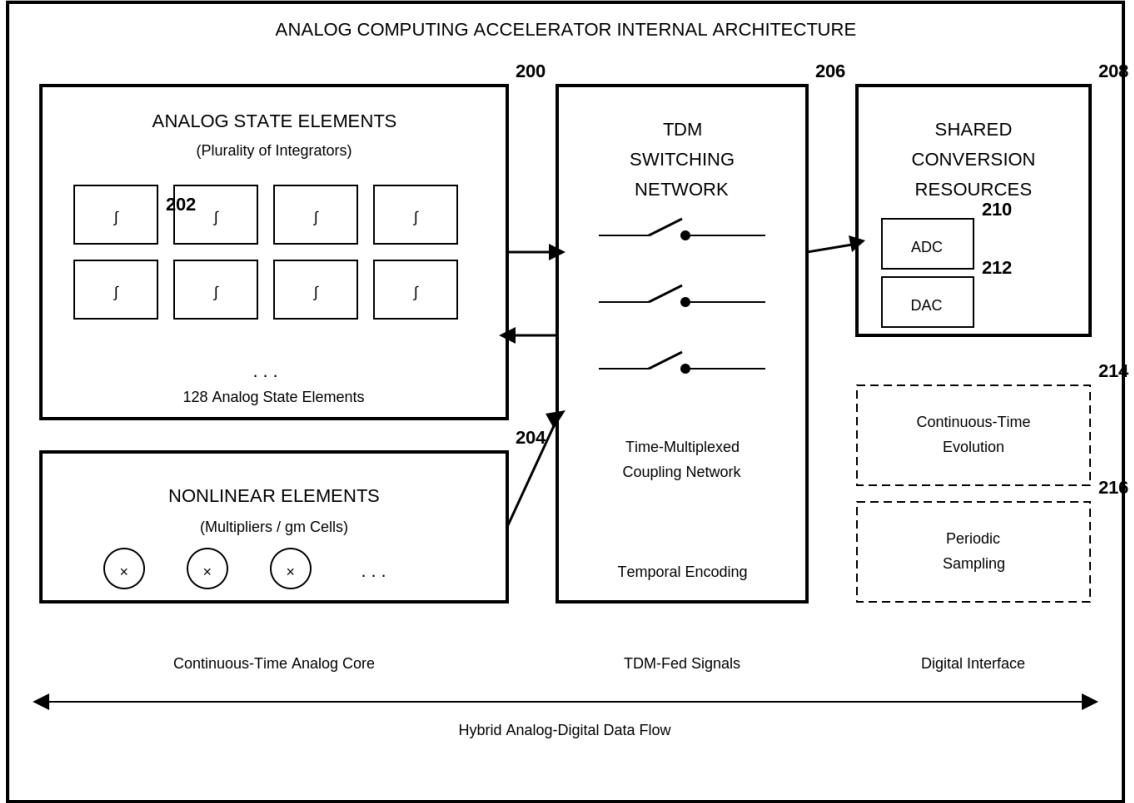


Figure 2: Internal Architecture

3.1 Description of Figure 2

Figure 2 depicts the internal architecture of the analog computing accelerator, showing the organization of computation resources.

Reference numerals:

- **200** — Analog Computing Accelerator Internal Architecture
- **202** — Analog State Elements (Plurality of Integrators) — 128 elements
- **204** — Nonlinear Elements (Multipliers / gm Cells)
- **206** — TDM Switching Network (Time-Multiplexed Coupling Network)
- **208** — Shared Conversion Resources
- **210** — ADC
- **212** — DAC
- **214** — Continuous-Time Evolution
- **216** — Periodic Sampling

The continuous-time analog core comprises analog state elements (202), shown as a plurality of

integrators, and nonlinear elements (204) such as multipliers and transconductance cells. The TDM switching network (206) provides temporal encoding of signal connections. Shared conversion resources (208) include ADC (210) and DAC (212) for interfacing with the digital domain. The hybrid analog-digital data flow supports both continuous-time evolution (214) and periodic sampling (216).

4 Figure 3 — Time-Division Multiplexed Analog Operation

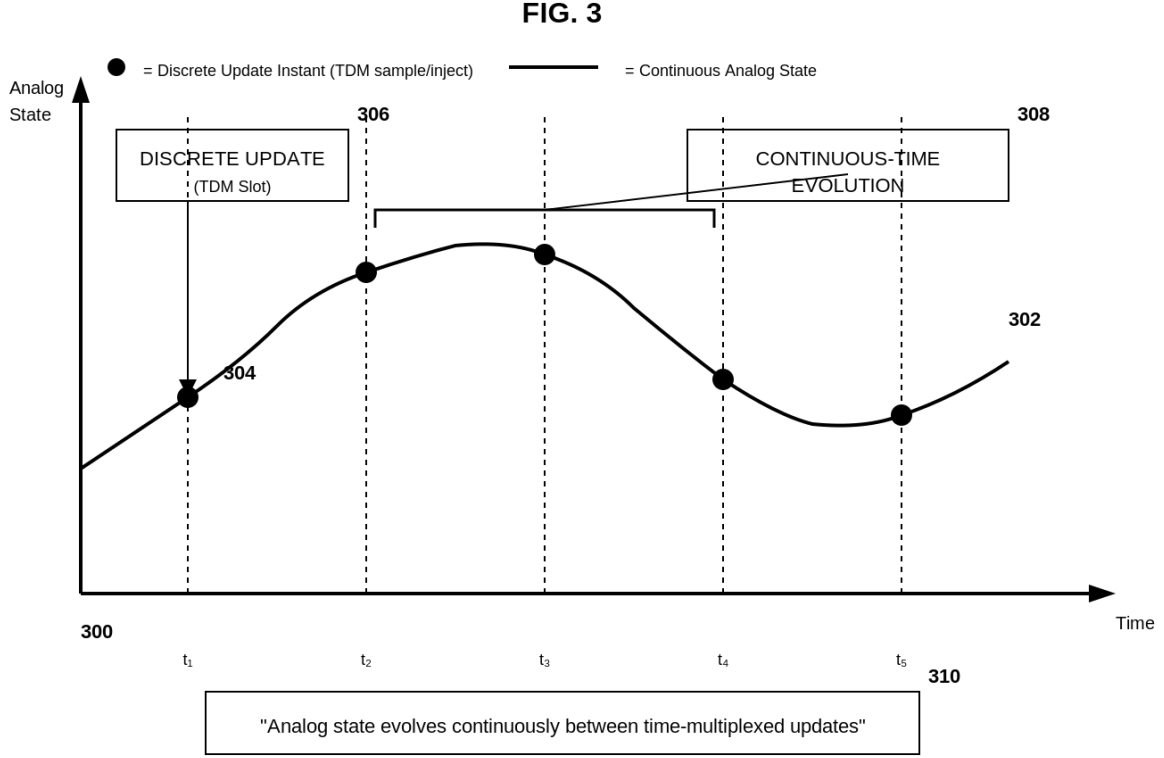


Figure 3: TDM Timing Diagram

4.1 Description of Figure 3

Figure 3 illustrates the timing relationship between discrete TDM updates and continuous analog state evolution.

Reference numerals:

- **300** — Time axis origin
- **302** — Continuous analog state trajectory
- **304** — Discrete update instant (TDM sample/inject) at t_1
- **306** — Discrete update (TDM Slot)
- **308** — Continuous-time evolution region
- **310** — Time axis (t_1, t_2, t_3, t_4, t_5)

The diagram shows that analog state evolves continuously between time-multiplexed updates. The filled circles represent discrete update instants (TDM sample/inject operations) occurring at times t_1 through t_5 . Between these discrete instants, the analog state (302) follows a continuous trajectory governed by the physical dynamics of the analog computation fabric. This demonstrates that computation within each allocated time slot is performed in the analog domain as a continuous physical process.

5 Figure 4 — TDM Coupling of Multiple Signals

FIG. 4

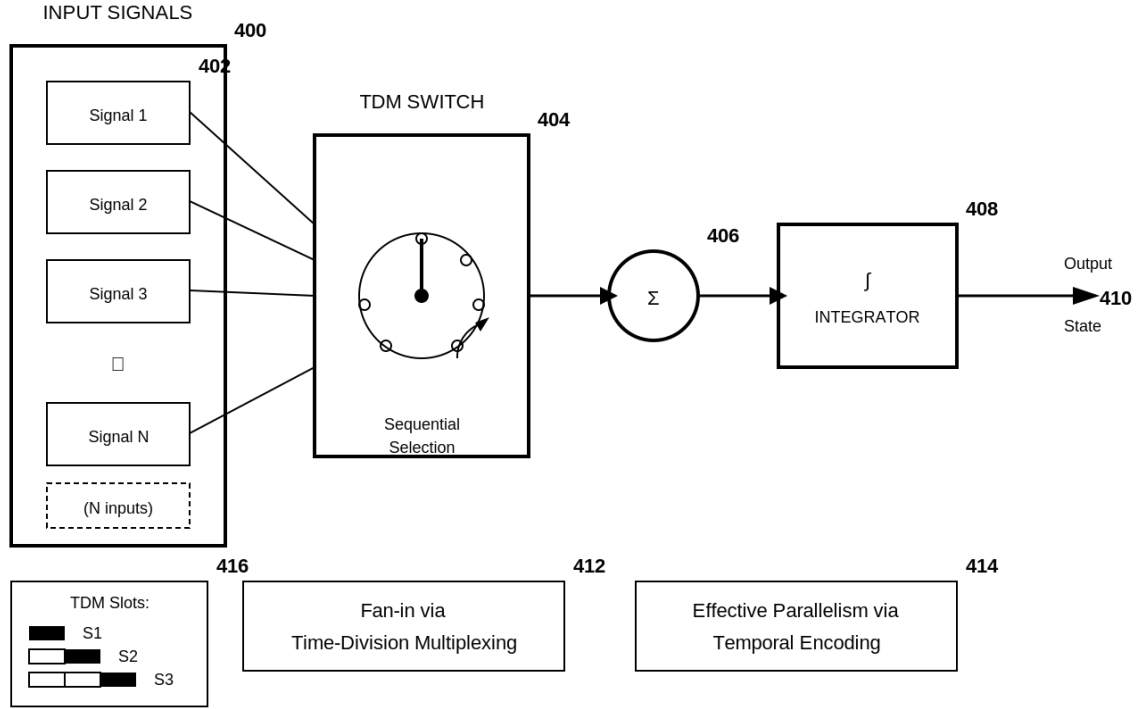


Figure 4: TDM Fan-in Architecture

5.1 Description of Figure 4

Figure 4 illustrates how time-division multiplexing enables multiple input signals to be coupled into a single analog computation element.

Reference numerals:

- 400 — Input signals section
- 402 — Signal 1 through Signal N (N inputs)
- 404 — TDM Switch (Sequential Selection)
- 406 — Summing node (Σ)
- 408 — Integrator ($\int$)
- 410 — Output State
- 412 — Fan-in via Time-Division Multiplexing
- 414 — Effective Parallelism via Temporal Encoding
- 416 — TDM Slots (S1, S2, S3...)

Multiple input signals (402) are sequentially selected by the TDM switch (404). The selected signals pass through a summing node (406) and are integrated by the integrator (408) to produce the output state (410). This architecture achieves fan-in via time-division multiplexing (412), enabling

effective parallelism via temporal encoding (414). The TDM slots (416) show how different time intervals are allocated to different input signals.

6 Figure 5 — Calibration and Measurement Loop

FIG. 5

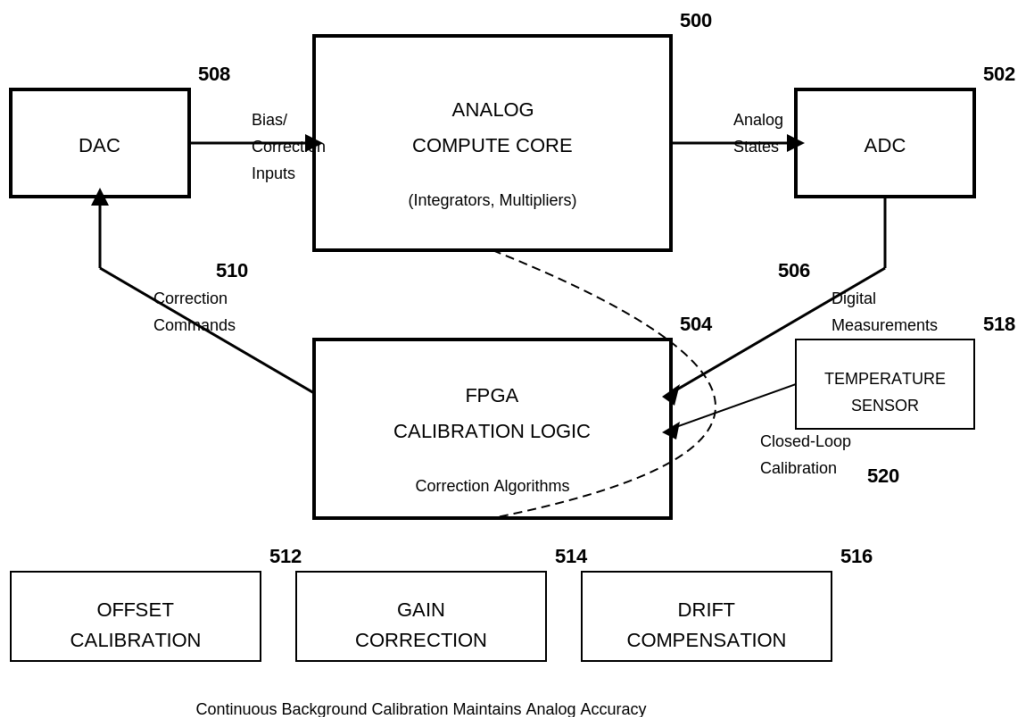


Figure 5: Calibration Feedback Loop

6.1 Description of Figure 5

Figure 5 depicts the closed-loop calibration system that maintains analog computation accuracy.

Reference numerals:

- **500** — Analog Compute Core (Integrators, Multipliers)
- **502** — ADC (Analog states measurement)
- **504** — FPGA Calibration Logic (Correction Algorithms)
- **506** — Digital Measurements path
- **508** — DAC (Bias/Correction Inputs)
- **510** — Correction Commands path
- **512** — Offset Calibration
- **514** — Gain Correction
- **516** — Drift Compensation
- **518** — Temperature Sensor
- **520** — Closed-Loop Calibration

The analog compute core (500) outputs analog states to the ADC (502), which provides digital measurements (506) to the FPGA calibration logic (504). The calibration logic implements cor-

rection algorithms including offset calibration (512), gain correction (514), and drift compensation (516). A temperature sensor (518) provides environmental data for temperature-indexed correction. Correction commands (510) are sent to the DAC (508), which applies bias and correction inputs back to the analog compute core. This forms a closed-loop calibration system (520) that continuously maintains analog accuracy.

7 Figure 6 — Software Control Interface

FIG. 6

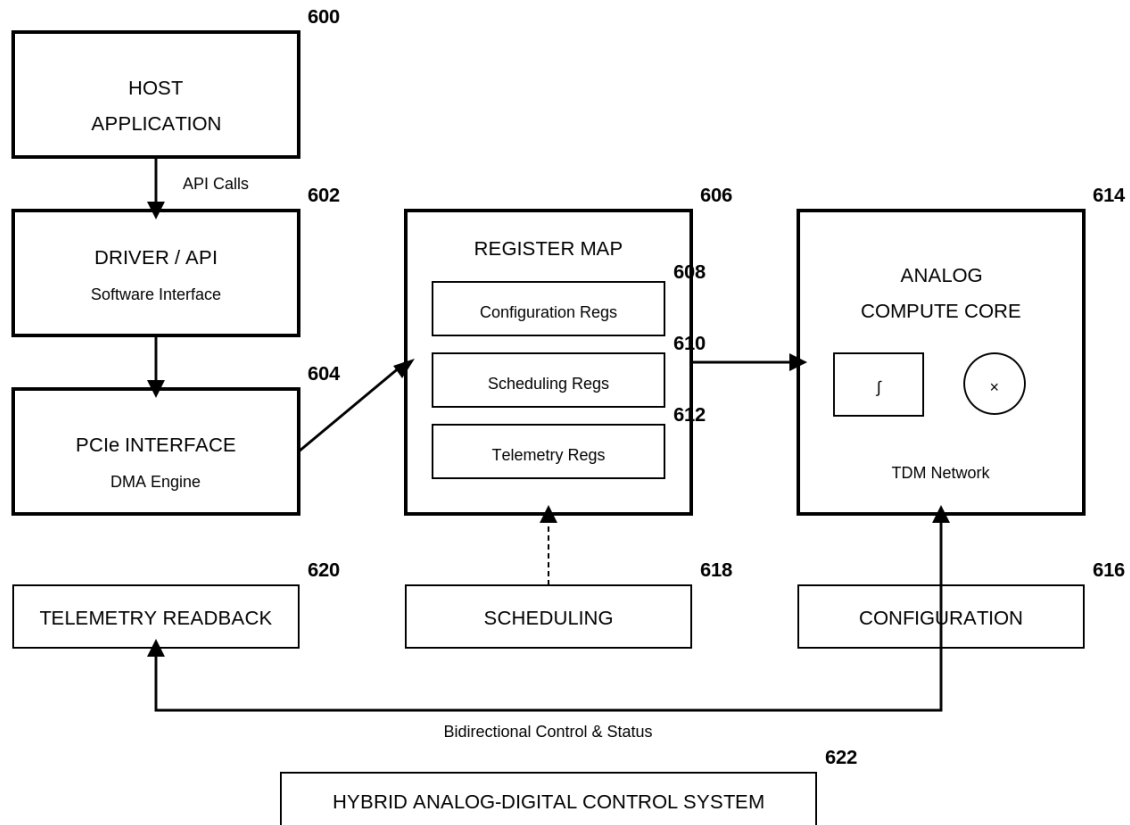


Figure 6: Software Control Interface

7.1 Description of Figure 6

Figure 6 illustrates the software interface architecture enabling host control of the hybrid analog-digital system.

Reference numerals:

- **600** — Host Application
- **602** — Driver / API (Software Interface)
- **604** — PCIe Interface (DMA Engine)
- **606** — Register Map
- **608** — Configuration Registers
- **610** — Scheduling Registers
- **612** — Telemetry Registers
- **614** — Analog Compute Core (with TDM Network)
- **616** — Configuration path
- **618** — Scheduling path
- **620** — Telemetry Readback path

- **622** — Hybrid Analog-Digital Control System

The host application (600) communicates through a driver/API layer (602) via API calls. The PCIe interface (604) provides DMA-based data transfer to the register map (606). The register map contains configuration registers (608), scheduling registers (610), and telemetry registers (612). These registers control the analog compute core (614) through configuration (616), scheduling (618), and telemetry readback (620) paths. The complete system forms a hybrid analog-digital control system (622) with bidirectional control and status capabilities.